Games People Play...

Max: Hmmm..The penny drops. You are not Job Job is not given to quoting scripture in his communications...So you are something of a paradox.
Hunt: Well, that depends.
Max: On what?
Hunt: Whether you like a paradox - I want $150,000 today.

- Mission Impossible

HAIMLC601 Game Theory Lecture 13 Analyzing Games

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Outline

1. Motivation
2. Cournot's Model
3. Bertrand's Model
4. The Game of Nim
5. The Game of Nim

Market Structure

Motivation

With a large number of buyers:

- A **monopoly market** is where there is one seller
- A **duopoly market** is where there are two sellers
- An **oligopoly market** is where there are few sellers

Market Structure

	Number of firms	Similarity of goods
Perfect competition	Many	Identical
Monopolistic competition	Many	Distinct
Oligopoly	Few	Identical or distinct
Monopoly	One	Unique

COURNOT (NASH) EQUILIBRIUM

Definition

The Cournot model of oligopoly assumes that:

- Rival firms produce a homogenous product
- Each attempts to maximize profits by choosing how much to produce.
- All firms choose output (quantity) simultaneously.
- The basic Cournot assumption is that each firm chooses its quantity, taking as given the quantity of its rivals.

The resulting equilibrium is a Nash equilibrium in quantities, called a **Cournot (Nash) equilibrium.**

COURNOT (NASH) EQUILIBRIUM

The Cournot Model of Duopoly

- There are two competing firms producing a single homogeneous product.
- These firms must choose how much of the good to produce.

COURNOT (NASH) EQUILIBRIUM

The Cournot Model of Duopoly

- The cost of producing one unit of the good is a constant c, the same for both firms.
- If a Firm i produces the quantity q_i units of the good, then the cost to Firm i is cq_i, for $i = 1, 2$.

COURNOT (NASH) EQUILIBRIUM

The Cournot Model of Duopoly

- The price of a unit of the good is negatively related to the total amount produced.
- If Firm 1 produces q_1 and Firm 2 produces q_2 for a total of $Q = q_1 + q_2$, the price is

$$P(Q) = \begin{cases} a - Q & \text{if } 0 \leq Q \leq a \\ 0 & \text{if } Q > a \end{cases}$$

For some constant a

COURNOT (NASH) EQUILIBRIUM

The Cournot Model of Duopoly

- The payoffs for the two players are the profits
  $u_1(q_1, q_2) = q_1 P(q_1 + q_2) - cq_1 = q_1(a - q_1 - q_2) - cq_1$
  $u_2(q_1, q_2) = q_2 P(q_1 + q_2) - cq_2 = q_2(a - q_1 - q_2) - cq_2$
- Assumption: $c < a$

COURNOT (NASH) EQUILIBRIUM

In a Monopolistic case

- Suppose $q_2 = 0$
- Then the return to Firm 1 if it produces q_1 units is $u_1(q_1) = q_1(a - q_1) - cq_1$
- The firm will choose q_1 to maximize this quantity
- We have, $u_1(q_1) = q_1(a - c) - q_1^2$

COURNOT (NASH) EQUILIBRIUM

In a Monopolistic case

- $u_1(q_1) = q_1(a - c) - q_1^2$
- To find the point at which the maximum occurs by taking a derivative with respect to q_1, setting it to zero and solving for $q1$
  $u'(q_1) = a - c - 2q_1 = 0$
- whose solution is $q_1 = (a - c)/2$.
- The monopoly price is $P((a - c)/2) = (a + c)/2$
- And the monopoly profit is $u_1((a - c)/2) = (a - c)^2/4$.

COURNOT (NASH) EQUILIBRIUM

For a Duopoly Model

- We find simultaneously the value of q_1 that maximizes u_1 and the value of q_2 that maximizes u_2 by setting the partial derivatives to zero.
- Where
  $u_1(q_1, q_2) = q_1(a - q_1 - q_2) - cq_1$
  $u_2(q_1, q_2) = q_2(a - q_1 - q_2) - cq_2$
- Taking Partial Derivatives, (u_1 w.r.t q_1 and u_2 w.r.t q_2)
  $\frac{\partial}{\partial q_1} u_1(q_1, q_2) = a - 2q_1 - q_2 - c = 0$
  $\frac{\partial}{\partial q_2} u_2(q_1, q_2) = a - q_1 - 2q_2 - c = 0$

COURNOT (NASH) EQUILIBRIUM

For a Duopoly Model

- Solving these equations simultaneously and denoting the result by q_1^* and q_2^*, we find
- $q_1^* = (a - c)/3$ and $q_2^* = (a - c)/3$
- In this Equilibrium, each firm produces less than the monopoly production, but the total produced is greater than the monopoly production.
- The payoff each player receives is

$$u_1(q_1^*, q_2^*) = \frac{a-c}{3}(a - \frac{a-c}{3} - \frac{a-c}{3}) - c\frac{a-c}{3} = \frac{(a-c)^2}{9}.$$

- The duopoly price is $P(q_1^* + q_2^*) = (a + 2c)/3$

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Cournot Model

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Cournot Model

Quantity Setters

- Describe how oligopolist firms that choose quantities can be modeled using game theory.
- The Cournot model considers firms that make an identical product and make output decisions simultaneously.

Bertrand Model

Price Setters

- Describe how oligopolist firms that choose prices can be modeled using game theory.
- The Bertrand model considers firms that make an identical product but compete on price and make their pricing decisions simultaneously.

Bertrand Model

Price Setters

- If each firm charges the price of c then the other firm can do no better than charge the price of c also
- If it raises its price it sells no output, while if it lowers its price is makes a loss
- (c, c) is a Nash equilibrium.

Bertrand Model

Price Setters

No other pair (p_1, p_2) is a Nash equilibrium since

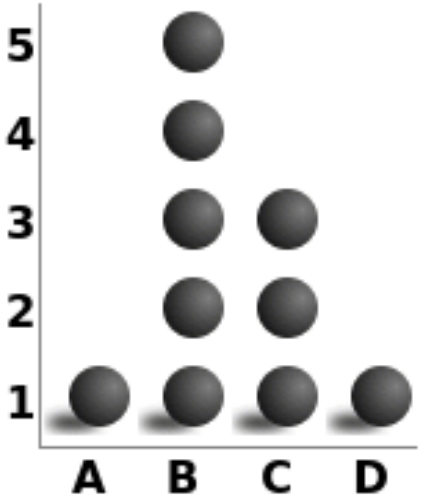
- If $p_i < c$ then the firm whose price is lowest can increase its profit by raising its price to c
- If $p_i = c$ and $p_j > c$ then firm i is better off increasing its price slightly
- if $p_i \geq p_j > c$ then firm i can increase its profit by lowering p_i to some price between c and p_j

Comparison of Models

Comparison

- **Bertrand's model:** Firm changes its behavior if it can increase its profit by changing its price, on the assumption that the other firm will not change its price but the other firm's output will adjust to clear the market.
- **Cournot's model:** Firm changes its behavior if it can increase its profit by changing its output, on the assumption that the output of the other firm will not change but the price will adjust to clear the market.

The Game of Nim



The Game of Nim

Another Version

- **Subtraction:**
- Starting with a pile of $x \in N$ chips.
- Two players alternate taking 1 to 4 chips.
- The player who removes the last set of chip wins.

The Game of Nim

Observation

Starting with:	First Player	Second Player
$x \leq 4$	Wins	
$x = 5$		Wins
$x = 6, .., , 9$	Wins	
$x = 10$		Wins

The Game of Nim

Observation

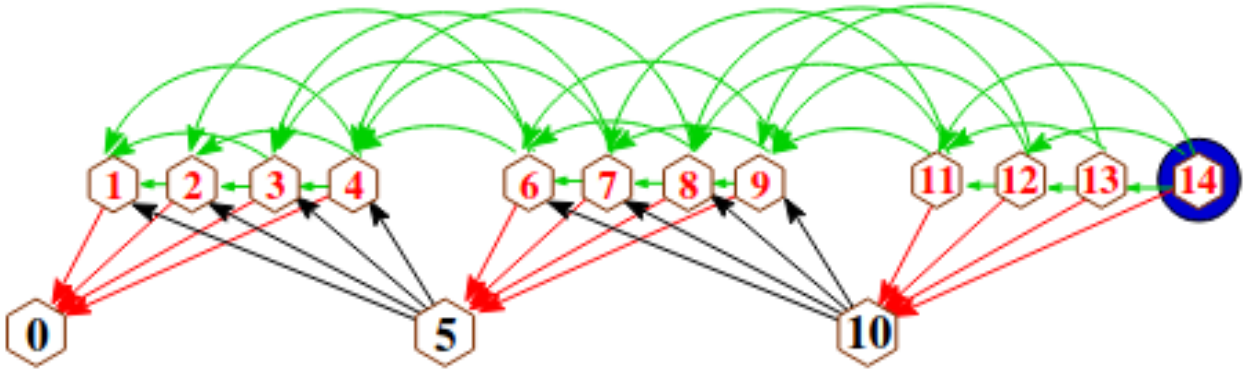
- $N = \{x \in \mathbb{N} |$ the first ("next") player can ensure a win if there are x chips at the start $\}$
- $P = \{x \in \mathbb{N} |$ the second ("previous") player can ensure a win if there are x chips at the start $\}$

The Game of Nim

Observation

- $\{1, 2, 3, 4, 6, 7, 8, 9\} \subseteq N$
- $\{0, 5\} \subseteq P$
- That is,
- $P = \{x | x \ \ mod \ \ 5 = 0\}$
- $N = \{x | x \ \ mod \ \ 5 \neq 0\}$

The Game of Nim



The Game of Nim

Observation

- $(1,0) \in N$ and $(0,1) \in N$
- $(1,1) \in P$
- $(1,2) \in N$ and $(2,1) \in N$
- $(1,2,3) \in P$
- $(1,2,3,4) \in N$

The Game of Nim

Bouton's solution of Nim

Nim Sum

	decimal	binary
x_1	3	0011
x_2	9	1001
x_3	13	1101
$x_1 \oplus x_2 \oplus x_3$	7	0111

The Game of Nim

Bouton's solution of Nim

A Nim position $x = (x_1, x_2, \ldots, x_k)$ is in P if and only if the Nim-sum of its components is 0

The Game of Nim

Bouton's solution of Nim

decimal	binary
1	0 1
2	1 0
3	1 1
0	0 0

The Game of Nim

Bouton's solution of Nim

	decimal	binary
x_1	6	0 1 1 0
x_2	12	1 1 0 0
x_3	15	1 1 1 1
$s = x_1 \oplus x_2 \oplus x_3$	5	0 1 0 1

The Game of Nim

Bouton's solution of Nim

	decimal	binary
x_1	6	0 1 1 0
x_2	12	1 1 0 0
x_3	15	1 1 1 1
$s = x_1 \oplus x_2 \oplus x_3$	5	0 1 0 1

The Game of Nim

Exercise

- 4, 5, 8
- 7, 9, 7, 4
- 2, 3, 6, 9, 7